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COURSE: Data Visualization

COURSE INSTRUCTOR: Dr. Ihtisham Ali

LAB INSTRUCTOR: Mr. Jawad Safdar

TITLE: Project Proposal

DEPARTMENT: IT and CS

PROGRAM: BS DS F23

Dataset to be Used (Source/Link)

The dataset used in this project is the *Sample Superstore Dataset*, which contains detailed information about sales, profit, customers, regions, and product categories.

Source: Kaggle

Link: <https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>

Problem Statement

Organizations often face difficulty in analyzing large volumes of sales data to identify patterns, profitability, and areas of improvement. Without proper visualization, decision-making becomes inefficient. This project focuses on transforming raw sales data into meaningful insights using data visualization techniques.

Objectives of the Project

- Analyze overall sales and profit performance
 - Identify loss-making products and regions
 - Understand customer segments and their behavior
 - Track sales trends over time
 - Develop an interactive dashboard for business insights
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Target Audience / Stakeholders

- Business Managers
 - Sales Teams
 - Marketing Analysts
 - Senior Executives
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Key Questions to be Answered through Visualization

- Which region generates the highest sales and profit?
- Which products or categories are underperforming?
- What are the monthly and yearly sales trends?
- Which customer segment contributes the most revenue?

- How do discounts impact profit?
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Data Cleaning & Preparation Plan

- Handle missing/null values
 - Remove duplicate records
 - Convert data types (e.g., date fields)
 - Standardize column names
 - Create calculated fields (Profit Ratio, Discount Impact)
 - Filter irrelevant or inconsistent entries
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Storyboard / Dashboard Layout Plan

- **Top Section:** KPIs (Total Sales, Total Profit, Total Orders)
 - **Middle Section:**
 - Sales by Region (Bar/Map)
 - Profit by Category (Bar Chart)
 - **Bottom Section:**
 - Sales Trend Over Time (Line Chart)
 - Sub-category Analysis (Tree Map)
 - **Side Filters:** Region, Category, Segment
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Tools to be Used

- Microsoft Power BI for dashboard development
 - Microsoft Excel for initial exploration
 - Python for preprocessing
 - SQL for data extraction and querying
 - Gamma AI for making a presentation
 - Microsoft Word for documentation purposes
 - GitHub for storing a project by making a repository.
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Workflow

Python Workflow

- Import dataset using pandas
- Perform data cleaning (null handling, duplicates removal)
- Convert date columns and create new features
- Perform exploratory data analysis (EDA)
- Export cleaned dataset for visualization

SQL Workflow

- Import dataset into database
- Use SELECT queries for filtering and aggregation
- Apply GROUP BY for category/region analysis
- Use JOINS (if multiple tables exist)
- Generate summarized tables for reporting

Power BI Workflow

- Load cleaned dataset
- Create relationships (if needed)
- Build calculated measures (DAX)
- Design interactive visuals (charts, maps, KPIs)
- Add filters and slicers for user interaction
- Publish dashboard

Real-World Application

This project enables organizations to make data-driven decisions by identifying profitable areas, reducing losses, and optimizing operations. It helps improve sales strategies, customer targeting, and overall business performance.

This project demonstrates a complete industry-level data visualization workflow, covering data extraction, cleaning, analysis, and interactive dashboard development for effective decision-making

Project Timeline:

Date Range	Task
April 12 – April 14	Data collection and dataset understanding
April 15 – April 18	Data cleaning and preprocessing
April 19 – April 22	Exploratory data analysis and insights
April 23 – April 26	Dashboard development in Power BI
April 27 – April 30	Final testing, refinement, and presentation

Visual Flow:

